# Lukas William Mayer

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## Education

**PhD Cognitive Science** — **2028 (Expected)**
*University of California, Irvine* — *Irvine, California, USA*

**Thesis:** Bayesian modelling of inversion problems in human-AI interaction (preliminary title)
**Advisor:** Professor Mark Steyvers
**Focus:** Computational modelling of human problem-solving abilities and alignment of AI systems with human cognition

**MSc Behavioral and Data Science (Distinction)** — **2022**
*University of Warwick* — *Coventry, England, UK*

**Thesis:** Cognition-Inspired Heuristics Improve Machine-Learning Predictions in Multi-Attribute Multi-Alternative Choice
**Advisor:** Professor Nick Chater
**Awards: Best Overall Score**: Behavioral and Data Science (equivalent to **Summa Cum Laude**)
**Focus:** Unifying models from computational cognitive science and machine learning

**BSc with Honors Psychology (First-Class)** — **2021**
*University of Stirling* — *Stirling, Scotland, UK*

**Thesis:** Cooperation does not improve selective and/or sustained attention
**Advisors:** Dr. John Michael and Dr. Ross Goutcher
**Awards: Peter McEwan Prize** for the Best Single Honours Psychology Student (equivalent to **Summa Cum Laude**)
**Alan Baddeley Prize** for the Best Psychology Dissertation
**Focus:** Measuring cognitive abilities in online games

## Research Experience

**Research Assistant: Center for Research and Evidence on Security Threats** — **2022 - 2023**
*University of Bath, School of Management* — *Bath, England, UK*

Conducted large-scale mixed methods study concerned with the prediction of online group communications which translate into real-world actions
Developed own project investigating algorithms for complex research appointment scheduling
**Principal Investigator**: Professor Adam Joinson

**Research Assistant: Centre for Cognition in Complex Environments** — **2019 - 2021**
*University of Stirling, Department of Psychology* — *Stirling, Scotland, UK*

Execution and data analysis of experiments concerning face matching and concealed identity recognition
**Principal Investigator**: Dr. Ailsa E. Millen

**Research Assistant: Centre for Cognition in Complex Environments** — **2021**
*University of Stirling, Department of Psychology* — *Stirling, Scotland, UK*

Automated video analysis for discrimination learning in lumpfish
**Principal Investigator**: Dr. Jamie Murray

## Publications

Karny, S., **Mayer, L. W.**, Ayoub, J., Song, M., Su, H., Tian, D., Moradi-Pari, E., & Steyvers, M. (2024). Learning with AI Assistance: A Path to Better Task Performance or Dependence? *Proceedings of the ACM Collective Intelligence Conference, 10–17.* https://doi.org/10.1145/3643562.3672610

Steyvers, M., Tejeda, H., Kumar, A., Belem, C., Karny, S., Hu, X., **Mayer, L. W.**, & Smyth, P. (2024). The Calibration Gap between Model and Human Confidence in Large Language Models. arXiv preprint arXiv:2401.13835.

**Mayer, L. W.**, Bocheva, D., Hinds, J., Brown, O., Piwek, L., Ellis, D. (under Review) Waste not want not: Computational methods to maximise attendance in group research. *Behavior Research Methods.*

## Miscellaneous

**Digital Methods Initiative Winter School - The use and misuse of Open Source Intelligence** **2023**
*University of Amsterdam, Amsterdam, North Holland, The Netherlands*

**International Interdisciplinary Computational Cognitive Science Summer School** **2023**
*University of Tübingen, Tübingen, Baden-Württemberg, Germany*

**NeuroMatch Academy Summer School: Deep Learning** **2022**
*NeuroMatch Academy, Los Angeles, California, USA*